

Money Saving Algorithm:
Reducing Energy Bill Costs With a Smart Battery
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Abstract:

This study evaluates the performance of a daily threshold-prediction algorithm designed to reduce Occidental College's monthly facility charge through battery-assisted peak shaving. The algorithm predicts each day's discharge threshold by applying a linear correlation between previous-day energy usage and historically optimal thresholds. Training on 2019–2023 load data produced a coefficient of determination of $R^2 = 0.72$, indicating a strong positive correlation and supporting the viability of threshold prediction from prior-day consumption. To assess performance of the model, a year-long simulation was conducted using a 2 MWh, 1 MW battery system controlled by the algorithm's threshold decisions. The model autonomously charged and discharged the battery to minimize daily peak demand, resulting in measurable reductions to both the monthly peak and overall demand profile. Across 2024, the algorithm achieved total cost savings of \$13,049, including \$13,700 from reduced facility charges, -\$17 from low-period peak reduction, and -\$308 from high-period peak reduction.

Introduction:

The rapid electrification that we are seeing in our cities is increasing overall electricity demand at a rapid pace. At the same time, the integration of renewable energy sources such as solar and wind is creating variability in the electricity supply. During periods of low or moderate demand, a significant portion of electricity can be supplied by renewables, but, during times of high demand, utilities must rely on fast-ramping, carbon-intensive natural gas *peaker plants*. These plants exist solely to cover short-duration spikes and are both the most expensive and most polluting sources on the grid. As overall electricity use continues to grow, these peaks become more costly and environmentally damaging. Battery storage can help by storing energy when demand is low or renewable output is high and discharging during peak periods, reducing both costs and emissions.

However, the economic value of a battery depends critically on how intelligently it is operated. Suboptimal control strategies can waste stored energy, miss peak events, or provide little benefit relative to the cost of the system. Optimizing battery operation makes these systems more economically viable and increases the potential to use renewable energy efficiently.

Occidental College faces similar challenges. The college experiences sharp daily peaks that contribute disproportionately to monthly electricity costs, particularly the *facility charge*, which is based on the highest instantaneous power demand in the previous 12 months and typically accounts for about 20% of the monthly bill. These peaks also result in greater reliance on carbon-intensive energy sources from the grid. By strategically discharging stored energy during high-demand periods, the college can lower its peak demand, reduce emissions, and save money. Other important charges in calculating savings are the high-period demand charge and the low-period demand charge. These are monthly fees based on the highest power peak that occurs during certain *stress* hours on the grid. The high-period covers the middle of the afternoon, when grid demand is typically highest, and the low-period covers the late morning and early evening. Both charges increase significantly during the summer months when electricity use across the region is at its peak.

This project investigates a data-driven, predictive approach to battery-based *peak shaving*. In this context, the *optimal threshold* is the battery discharge level that limits how much energy is drawn from the grid on a given day. Any demand above this threshold is met by the battery, helping to reduce short-duration peaks that contribute disproportionately to the monthly facility charge. This in effect will level, or shave, off the highest energy demands of each day. Using historical campus energy usage from 2019–2023, a correlation was found between each day's total energy consumption and that day's optimal threshold. A fitted line from this correlation is then used to predict the threshold for the next day. The algorithm applies this predicted threshold in 2024, to decide when the battery should charge or discharge, with the goal of reducing facility charges and overall electricity costs.

The primary goal of this project is to determine how much money Occidental College could save by using a predictive, battery-based peak-shaving algorithm. Comparing the cost of the battery to annual estimated cost reductions, the study evaluates whether the simulated battery system would be a good investment under current energy usage patterns and electricity rates and charges.

Theory:

The focus of this project was on reducing the facility charge first, because it carries the highest cost per unit of power on the campus energy bill. At Occidental College, the facility charge is 8.95 per kW per month and typically accounts for around 20% of the total monthly bill. This high cost per kilowatt makes it a priority target for cost reduction, as even modest decreases in peak demand can translate into substantial savings.

To minimize the facility charge, an *optimal threshold* for each day was calculated. The optimal threshold represents the maximum level of power that can be drawn from the grid while still allowing the battery to cover all demand above that threshold. This ensures the battery fully utilizes its capacity to shave the daily peak. Mathematically, the threshold was determined by integrating the portion of the daily power profile above the threshold, P_{thresh} , and adjusting P_{thresh} until the integral equaled the battery's total energy capacity:

$$E_{Battery} = \int_{t_0}^{t_f} [P(t) - P_{thresh}]_+ dt$$

where $P(t)$ is the power demand at time t , $[\cdot]_+$ indicates only the portion above the threshold, and $E_{Battery}$ is the battery's energy capacity. This method ensures the battery reduces the peak as much as possible without running out of charge.

Once the optimal thresholds were calculated for historical days, the next step was to identify a predictive relationship between these thresholds and variables that could be measured in advance. Using a machine-learning approach based on correlation analysis, several candidate predictor variables were tested: temperature, previous-day peak

demand, and previous-day total energy use. The goal was to find the most linearly correlated variable to the optimal threshold. Among these, previous-day total energy consumption exhibited the highest correlation, with an R^2 value of 0.72, indicating a strong linear relationship.

A prediction model was then created by fitting a line through the points representing the optimal threshold versus previous-day total energy use. The line was adjusted to pass slightly above most points to avoid setting the threshold too low, which could cause the battery to run out of charge prematurely. If the battery was depleted before the end of the peak, the remaining demand would be drawn from the grid at full power, negating potential savings. Using this predictive line, the daily threshold for battery discharge could be estimated in advance, guiding battery operation to maximize both peak shaving and cost reduction.

Data:

Table 1: 2024 Original Energy Bill Costs

Month	High Peak (MW)	Low Peak (MW)	Low Demand Cost (USD)	High Demand Cost (USD)	Total Demand Cost (USD)	Monthly facility charge (USD)	Monthly Total (USD)
1	1.64	1.69	0	7073	7073	27701	41846
2	1.68	1.80	0	7232	7232	27701	42165
3	1.66	1.70	0	7133	7133	27701	41967
4	1.77	1.72	0	7601	7601	27701	42903
5	1.67	1.75	0	7199	7199	27701	42099
6	2.13	2.09	6881	20660	27541	27701	82783
7	2.27	2.21	7295	22051	29346	27701	86392
8	2.23	2.22	7316	21616	28932	27701	85565
9	2.65	2.81	9272	25738	35010	27701	97721
10	2.31	2.64	0	9913	9913	27701	47526
11	1.66	1.93	0	7139	7139	27701	41978
12	1.65	1.70	0	7111	7111	27701	41923
		Yearly Totals	Sum Low Peak	Sum High Peak	Sum Demand Charge	Total facility charge	Total Year Cost
			30763	150467	181230	332407	694867

Table 2: 2024 Reduced Energy Bill Costs

Month	High Peak (MW)	Low Peak (MW)	Low Demand Cost (USD)	High Demand Cost (USD)	Total Demand Cost (USD)	Monthly facility charge (USD)	Monthly Total (USD)
1	1.64	1.69	0	7073	7073	26559	40704
2	1.68	1.89	0	7232	7232	26559	41023
3	1.66	1.84	0	7133	7133	26559	40825
4	1.76	1.94	0	7579	7579	26559	41718
5	1.67	1.93	0	7199	7199	26559	40957
6	2.09	2.09	6881	20275	27156	26559	80871
7	2.27	2.21	7295	22051	29346	26559	85250
8	2.23	2.30	7605	21616	29222	26559	85002
9	2.73	2.73	8999	26453	35452	26559	97464
10	2.31	2.60	0	9913	9913	26559	46384
11	1.66	1.93	0	7139	7139	26559	40836
12	1.65	1.85	0	7111	7111	26559	40781
		Yearly Totals	Sum Low Peak	Sum High Peak	Sum Demand Charge	Total facility charge	Total Year Cost
			30781	150775	181555	318708	681818

This table shows the original cost calculation for the facility charge and demands charges for 2024.

Table 3: 2024 Savings by Month and Charge

Month	Low Cost (USD)	Demand Saved	High Demand Cost (USD)	Demand Saved	Total Demand Cost (USD)	Demand Saved	Monthly facility charge (USD)	charge Saved	Monthly Total Saved (USD)
1	0		0		0		1142		1142
2	0		0		0		1142		1142
3	0		0		0		1142		1142
4	0		22		22		1142		1185
5	0		0		0		1142		1142
6	0		385		385		1142		1911
7	0		0		0		1142		1142
8	-289		0		-289		1142		563
9	272		-715		-442		1142		257
10	0		0		0		1142		1142
11	0		0		0		1142		1142
12	0		0		0		1142		1142
	Sum Low Peak Saved		Sum High Peak Saved		Sum Demand Charge Saved		Total facility charge Saved		Total Year Cost Saved
	-17		-308		-325		13700		13049

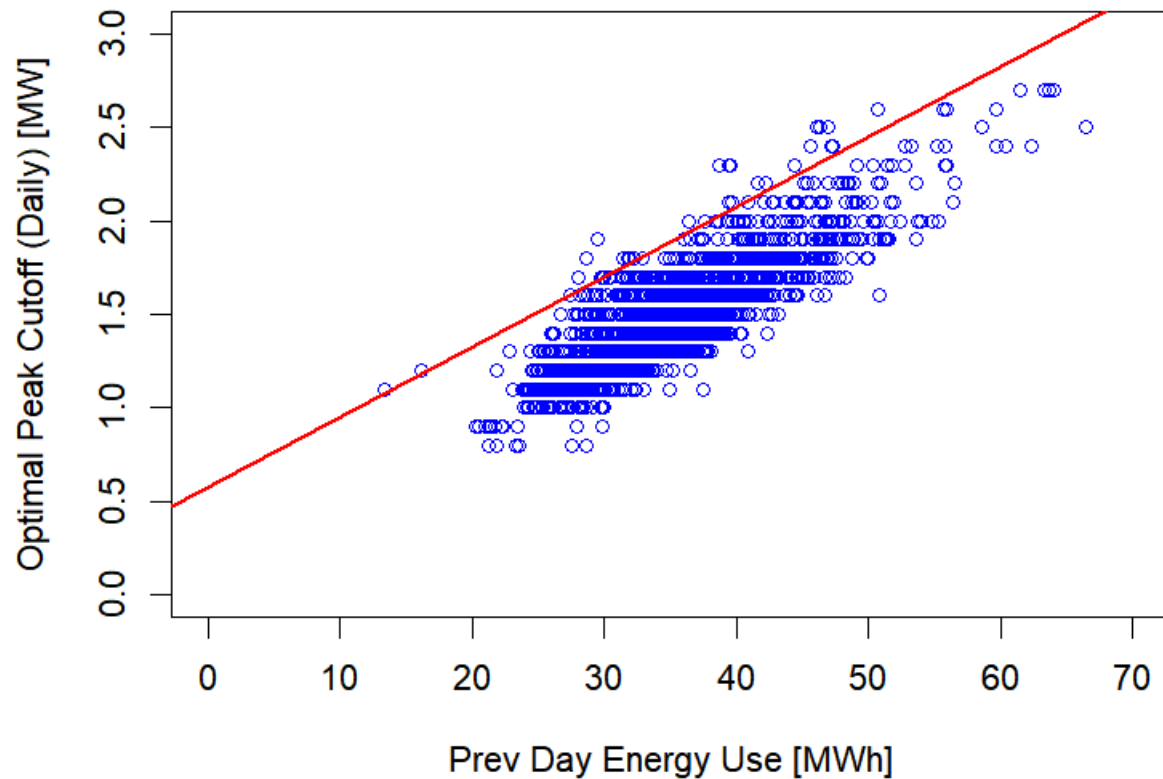
This table shows the reduced cost calculation for the facility charge and demand charges after simulating the effects of peak shaving algorithm and the battery.

Table 4: Pricing Plan by Each Charge Calculated

Charge Type	Summer (June - September)	Regular (November-May)
Facility (any hour)	\$8.95 per kW	\$8.95 per kW
High-Period Demand (1 pm to 5 pm)	\$9.70	\$4.30
Low-Period Demand (10 am to 1 pm and 5 pm to 8pm)	\$3.30 per kW	\$0 per kW

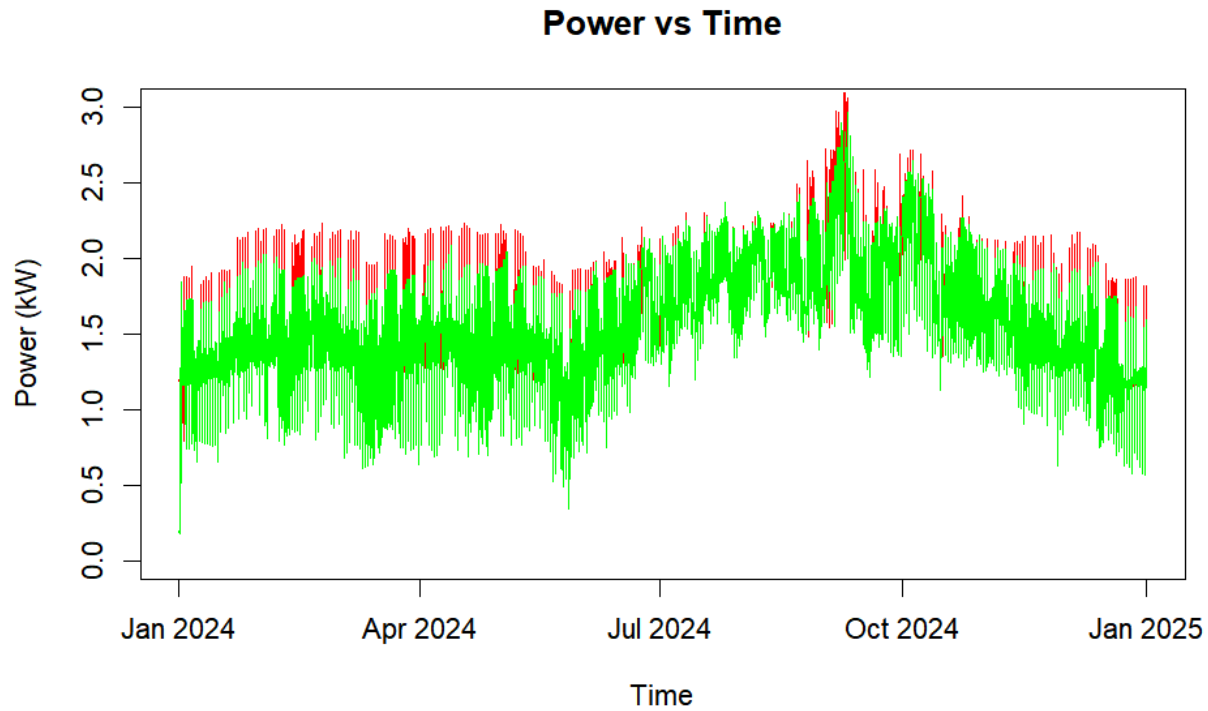
This table shows the pricing breakdown by hour and month for each charge used to calculate the net savings.

Plot 1: 2019 - 2023 Training Data and Fitted Prediction Model



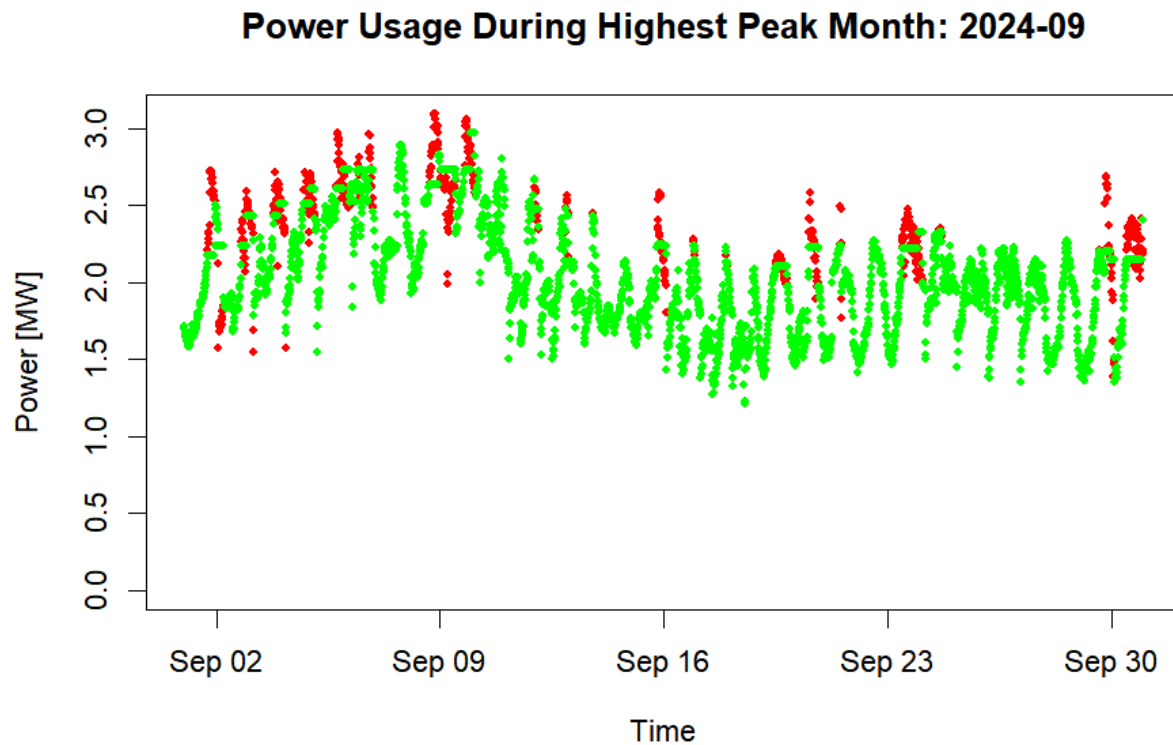
This graph shows how the model was trained on after plotting the previous day's energy use versus the optimal power cutoff. The model is defined by the red line that passes through the data as it takes a measured previous day's energy use value in MWhs and outputs a threshold value in MWs.

Plot 2: 2024 Original Power Signal (RED) and Reduced Power Signal (GREEN)



This plot shows the difference between the reduced power demand vs time compared to the original power demand vs time for the year 2024. The Red is the original and the green is the reduced values. Any red peaks that go above a green peak show exactly how much the battery and algorithm was able to cut down the power consumption at that time.

Plot 3: 2024 Largest Peak Month, Original Power Signal (RED) and Reduced Power Signal (GREEN)



This plot highlights the power demand for the month with the largest peak of the year 2024: September. This month is extremely important because it decides what the monthly facility charge will be for the whole year. We can see the highest peak is around September 9th, and the overlay of the green dots shows how much the power consumption was reduced.

Analysis:

To develop a predictive threshold that could be applied in real time, we needed a method for estimating each day's *optimal discharge threshold* using only information known before the day begins. After computing the optimal threshold for every day from 2019–2023 using the integration procedure described in the Theory section, we tested several possible predictor variables: the previous day's peak power demand, the previous day's average temperature, and the previous day's total energy consumption. For each candidate variable, we created a scatter plot with the predictor variable on the x-axis and the calculated optimal threshold on the y-axis, producing one point per day across the

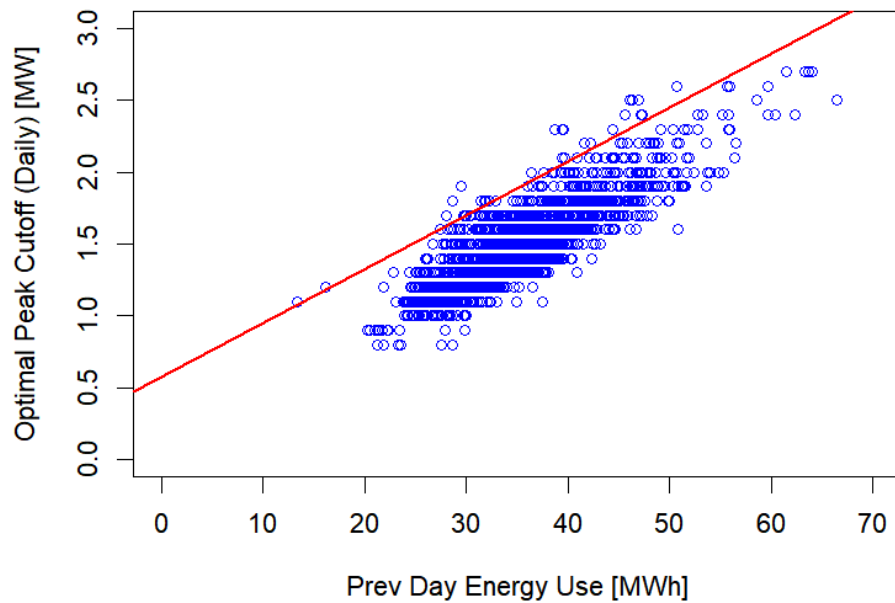
five-year dataset. These plots allowed us to visually assess whether a consistent relationship existed between the predictor and the optimal threshold.

For each candidate, we then performed a linear regression and computed the corresponding R^2 value. The objective was to find a variable with a strong, stable linear correlation so that a simple fitted line could serve as a reliable daily threshold predictor. Comparing the resulting regressions, the previous day's total energy consumption produced the strongest correlation, with the highest R^2 value among all variables. The $R^2 = 0.72$ indicates that roughly 72% of the daily variation in optimal thresholds can be explained by the previous day's total energy consumption. To reduce the risk of setting the threshold too low—an error that could cause the battery to discharge too early and allow a later peak to pass unshaved—the fitted line was chosen to lie slightly above most of the data points.

The final predictive model corresponds to the red regression line shown below. Its functional form is

$$f(E_{prev}) = 0.21692715 + (0.03753222) E_{prev}$$

where $f(E_{prev})$ returns the predicted daily discharge threshold in MW based on the previous day's total energy use E_{prev} (in MWh).



To evaluate the algorithm, the correlation-based model was applied to every day of the 2024 demand data. A simulation of the effect of the battery was run by instructing the battery to discharge whenever demand exceeded the threshold and charge whenever demand fell below it, subject to the battery's capacity limits. The resulting "battery-modified" demand signal was then run through the same billing structure that is used by the utility so that the facility and demand charges could be compared with and without the controlled battery. After testing the model on 2024 data, we simulated that the model would have saved a total of \$13,049 for the entire year. This is calculated by comparing the actual energy bill costs for the University to our simulated reduced costs and focusing on three specific charges on the bill: the High-Period Demand, the Low-Period Demand, and the facility charge. The breakdown for each of these charges is -\$17 for the Low-Period Demand charge, -\$308 for the High-Period Demand charge, and +\$13,700 for the facility charge. Low-period, high-period, and facility charges by each month are shown in Table 3 in the data section.

Discussion:

The model primarily targeted reducing the facility charge rather than the demand charge. Because the College's daily peak load typically occurs outside of demand-pricing hours, the model produced minimal reductions in demand-related costs. In fact, a flaw in the battery-charging logic caused the simulation to incur *higher* demand charges: the algorithm allowed the battery to charge whenever the total power draw fell below a threshold, without prohibiting charging during demand-pricing periods. This resulted in unnecessary demand spikes and a loss in overall savings.

The most immediate improvement is straightforward: implement explicit time-based restrictions that prevent the battery from charging during demand-pricing hours. This adjustment alone would eliminate the unintended increase in demand charges. Beyond this correction, greater savings could be achieved by adopting season-specific strategies. The facility-charge-focused model performs best when the annual peak load is likely to occur—typically June through September. During winter months, however, a model optimized for reducing demand charges may generate additional value. Using two

tailored control strategies, each aligned with seasonal load characteristics, would likely improve cost-effectiveness.

Even with room for improvement, the algorithm and battery were able to save the college over \$13,000 in a single year. With the agreement between Occidental College and the battery manufacturer Critical Loop, Occidental College would provide approximately \$100,000 in upfront capital for the equipment while retaining 60% of the annual savings generated by the battery-control algorithm. Assuming 2024 is an accurate average annual savings indicator we can assume Occidental will pocket \$7,829 per year and thus it would take the college 12.8 years to break even on the investment. This payback period is long but still a reasonable investment for the college. These numbers are even more promising when considering the potential improvements to the algorithm listed above that could shorten the pay back period further and make the investment even better.

Conclusion:

This project set out to evaluate whether a battery-control algorithm could meaningfully reduce Occidental College's utility costs by targeting facility and demand charges, and the results show that the approach is both technically promising and financially viable with key improvements. The model successfully demonstrated significant facility-charge reductions, and although an error in the charging logic led to unintended demand-charge penalties, this issue is easily corrected through time-based charging restrictions. With these corrections—along with seasonal model adjustments to better capture both summer peak events and winter demand-charge opportunities—the algorithm is positioned to produce reliable annual savings. Under the Critical Loop partnership structure, these savings translate to a payback period of roughly thirteen years for the College, placing the investment within a realistic range for long-lived energy-storage assets.

More broadly, this work highlights the increasing importance of intelligent load-management strategies as campuses and institutions navigate rising energy costs, shifting utility rate structures, and statewide decarbonization efforts. Even modest improvements to peak-shaving and demand shaping can compound into substantial long-term savings, especially as renewable penetration grows and grid volatility

increases. By refining and deploying models such as this one, Occidental College would not only reduce operational expenses but also strengthen its role as an early adopter of innovative, sustainability-focused energy technologies.